

Problem

An ac motor with impedance $\mathbf{Z_L} = 2 + j1.2 \, \Omega$ is supplied by a 220-V, 60-Hz source. (a) Find pf, P , and Q . (b) Determine the capacitor required to be connected in parallel with the motor so that the power factor is corrected to unity.

Step-by-step solution

Step 1 of 9

Calculate the current value of the current passing through the impedance $\mathbf{Z_L}$.

$$\begin{aligned} \mathbf{I} &= \frac{220}{2 + j1.2} \\ &= \frac{220}{2.33 \angle 30.96^\circ} \\ &= 94.42 \angle -30.96^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current $\mathbf{I}$ passing through the impedance $\mathbf{Z_L}$ is,

$$94.42 \angle -30.96^\circ \text{ A}.$$

Step 2 of 9

(a)

Calculate the value of the power factor pf.

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute 0° for θ_v and -30.96° for θ_i .

$$\begin{aligned} \text{pf} &= \cos(0 - (-30.96^\circ)) \\ &= \cos(30.96^\circ) \\ &= 0.8575 \end{aligned}$$

Therefore, the value of the power factor pf is 0.8575.

Step 3 of 9

Calculate the value of the average power P .

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Substitute 94.42 A for I_{rms} , 220 V for V_{rms} , 0° for θ_v and -30.96° for θ_i .

$$\begin{aligned} P &= (220)(94.42) \cos(0 - (-30.96)) \\ &= (20772.4)(0.8575) \\ &= 17.812 \text{ kW} \end{aligned}$$

Therefore, the value of the average power P is 17.812 kW.

Step 4 of 9

Calculate the value of the reactive power Q .

$$Q = V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)$$

Substitute 94.42 A for I_{rms} , 220 V for V_{rms} , 0° for θ_v and -30.96° for θ_i .

$$\begin{aligned} Q &= (220)(94.42)\sin(0 - (-30.96)) \\ &= (20772.4)\sin(30.96) \\ &= (20772.4)(0.5144) \\ &= 10.686 \text{ kVAR} \end{aligned}$$

Therefore, the value of the reactive power Q is 10.686 kVAR.

The power factor is lagging since reactive power is positive.

Step 5 of 9

(b)

Calculate the value of the power factor angle after the capacitor is connected across the load.

$$\begin{aligned} \theta_2 &= \cos^{-1}(1) \\ &= 0^\circ \end{aligned}$$

Therefore, the value of the power factor angle θ_2 is 0° .

Step 6 of 9

Calculate the value of the apparent power absorbed by the load after the capacitor is connected across the load.

$$S_2 = \frac{P_2}{\cos \theta_2}$$

Substitute 0° for θ_2 and 17.812 kW for P_2 since, no change is seen in the value of the real power.

$$\begin{aligned} S_2 &= \frac{17.812 \text{ k}}{\cos 0^\circ} \\ &= \frac{17.812 \text{ k}}{1} \\ &= 17.812 \text{ kVA} \end{aligned}$$

Therefore, the value of the apparent power S_2 is 17.812 kVA.

Step 7 of 9

Calculate the value of the reactive power Q_2 after the capacitor is connected across the load.

$$Q_2 = S_2 \sin \theta_2$$

Substitute 0° for θ_2 and 17.812 kVA for S_2 .

$$\begin{aligned} Q_2 &= (17.812 \text{ k}) \cdot \sin(0^\circ) \\ &= (17.812 \text{ k})(0) \\ &= 0 \text{ VAR} \end{aligned}$$

Therefore, the value of the reactive power Q_2 is 0 VAR.

Step 8 of 9

Calculate the value of the reactive power Q_c due to capacitor.

$$Q_c = Q_1 - Q_2$$

Substitute 0 VAR for Q_2 and 10.686 kVAR for Q_1 .

$$\begin{aligned} Q_c &= 10.686 \text{ k} - 0 \\ &= 10.686 \text{ kVAR} \end{aligned}$$

Therefore, the value of the reactive power due to capacitor, Q_c is 10.686 kVAR.

Step 9 of 9

Calculate the value of the capacitance C

$$C = \frac{Q_c}{\omega V_{\text{rms}}^2}$$

Substitute $2\pi f$ for ω .

$$C = \frac{Q_c}{2\pi f V_{\text{rms}}^2}$$

Substitute 10.686 kVAR for Q_c , 60 Hz for f and 220 V for V_{rms} .

$$\begin{aligned} C &= \frac{10.686 \text{ k}}{2\pi(60)(220)^2} \\ &= \frac{10.686 \text{ k}}{2\pi(60)(48400)} \\ &= \frac{10.686 \text{ k}}{18246.37 \text{ k}} \\ &= 585.65 \text{ } \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitor C is 585.65 μF .